

The Observer and the Cosmological Datum: On the Ontological Invariance of the Observing Domain

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Abstract

An observer generated within a rendering domain cannot leave that domain, and the reason is not distance. Changing an observer's position does not change the observer's domain. Instruments extend the range of physical inputs accessible within a domain; they are manufactured, calibrated and read by mechanisms belonging to that domain, and they therefore multiply the channels available inside it rather than producing an observer outside it. A spacecraft beyond the galaxy carries the domain with it.

The consequence for cosmology is specific and is developed here through two cases. Every measurement of the cosmic microwave background has been acquired from within the Solar System, and the separation of line-of-sight foregrounds from the cosmological signal is constrained by extensive internal cross-validation which cannot constitute an observationally external null test. The large-scale alignment anomalies remain undecided after two decades, and the structure of that indecision is what this paper analyses. The Hubble tension is a discrepancy of a different kind, between a model-dependent early-Universe inference and a late-Universe distance-ladder determination; the cross-constraints among distance ladder, background radiation, baryon acoustic oscillations, time-delay lensing and megamaser determinations are real, and they are all obtained within one domain.

A cosmological datum is not a bare quantity but a quantity conditioned on observer state, instrument and domain. No domain-internal operation uniquely decomposes it into an observer-independent remainder. Falsification conditions are stated.

Keywords: observer; rendering domain; cosmological datum; foreground separation; large-scale alignment anomalies; Hubble tension; observational limits; domain-relative observability; Copernican principle

1. What the Companion Papers Left Unasked

Three companion papers precede this one, and their results are taken as established here and are not re-derived. Together they establish that the observer is the existential precondition of physical theory, that the observer's discriminative capacity is grounded in radiation, and that observing systems sharing a common radiative expression constitute a rendering domain within which mutual rendering is coherent and across which it is impossible. Each is stated below in the form in which it will be used.

The first is that the observer is not a peripheral element of physical theory but its existential precondition. The construction of a reference frame requires an entity capable of measurement; the Lorentz transformation connecting two frames becomes an operation on empty sets when both observers are stripped of measurement capacity; and the Minkowski metric, whose invariance is conventionally taken as proof of observer-independent reality, presupposes the observer it claims to transcend. Physics has operated with an unexamined existential presupposition.

The second is that the observer's discriminative capacity is not an additional faculty possessed by certain material arrangements but is grounded in radiation, and that every known act of discrimination and registration — biological, instrumental, computational — is performed through electromagnetic interaction. The convergence is total and is not a contingency of terrestrial technology.

The third is that a rendering domain is the set of observing systems sharing a common physical expression of the radiative substrate, and that such systems are capable of mutual rendering while systems in different domains are not. Within a domain the collective rendering is coherent: all terrestrial observers render the same physical world, and the structural isomorphism of their spacetimes follows from common radiative origin and common mechanism. Across domains the incommensurability is structural rather than spatial. The boundary between domains is not a wall that could be crossed by travelling far enough or by building a more powerful telescope.

That third result carries an assumption which was correct for its purpose and which is examined here. In establishing that domains are mutually inaccessible, the interior of a domain was treated as unproblematic: observers inside a domain agree, their instruments agree, and the world they render is one world. The interior was the premise from which the exterior conclusion was drawn.

What was not asked is this. If observers, their instruments, and the structures they observe are all products of one rendering, what is the status of the data such observers obtain about the rendering itself? An observer inside a domain can move. Human observers have moved: from the surface of the Earth to orbit, from orbit to the outer Solar System. Instruments have improved by many orders of magnitude in sensitivity and have extended far beyond the range of unaided human perception. The question is whether any of this constitutes a change of domain, and therefore whether the accumulation of range and sensitivity converges on an observation that is not conditioned on the observing domain.

It does not. That is the result of this paper, and it is stated as a theorem in Chapter 6. Spatial displacement of an observer is not ontological displacement of the observing domain. The chapters between develop what is required to establish it and what follows from it in the one science where the question has practical consequences, since cosmology is the only discipline whose object is the domain as a whole.

Two clarifications belong here, because the argument is easily misread in two directions and both misreadings are refused at the outset.

The first is that nothing here places the Earth, the Solar System, or human observers at the centre of anything. An observer is the centre of the observer's own field of view; this is a geometric fact that holds for any observer at any position and follows from the finite speed of light. It is not a claim about the structure of the world. Rendering is not centred on observers, and observers are themselves rendered. The distinction between the centre of a field of view and the centre of a rendering is maintained throughout and is stated formally in Chapter 6.

The second is that the two cosmological cases examined in Chapters 4 and 5 are not offered as evidence for the position advanced here. They are its subject matter. An anomaly with several candidate explanations supports none of them, and the analysis given here does not depend on either case being resolved in any particular direction. What is analysed is the structure of the situation in which they are undecided.

2. The Domain-Conditioning of a Cosmological Datum

A cosmological datum is conventionally treated as a quantity belonging to the world: the temperature of the microwave background in a given direction, the expansion rate, the amplitude of primordial fluctuations. Measurement is treated as the retrieval of such a quantity, with instrumental effects and local contaminations understood as additions to be removed. The datum is what remains after the removal.

This treatment is not wrong as a description of practice, and the practice is extraordinarily successful. What it leaves implicit is the structure of what is removed and against what standard the removal is judged complete.

A cosmological datum has the form $D(O, I, G)$. The first argument is the observer's state: position, velocity, and the relations these bear to the structures being observed. The second is the instrument and the measurement procedure: response function, calibration chain, systematic model, and the theoretical assumptions embedded in the conversion of raw counts into physical quantities. The third is the domain: the physical conditions under which observation is possible at all, which the companion papers

established to be the conditions of a rendering, including the mechanism through which discrimination occurs and the class of physical inputs that mechanism can register.

The question this paper addresses can now be stated exactly. Does there exist an operation F , available within the domain, such that F applied to $D(O, I, G)$ yields a unique D^* independent of O ?

Three points about this formulation require statement, because each corresponds to a way the question is ordinarily dissolved rather than answered.

First, the question is not whether the observer's contribution can be estimated and corrected. It plainly can, and is, to remarkable precision. The dipole anisotropy of the microwave background is the clearest instance: it is attributed to the motion of the Solar System with respect to the radiation, its amplitude yields a velocity, and it is subtracted as a matter of routine. The question is whether the correction is uniquely determined by domain-internal operations, or whether it is determined by a model of the observer's contribution which is itself constructed within the domain and validated against other domain-internal data.

Second, the question is not whether independent measurements exist. They exist in large numbers, and cosmology's practice of cross-validating results obtained by unrelated methods is one of its principal strengths. Different instruments, frequencies, sky regions, epochs, and analysis pipelines produce results that constrain one another tightly. What must be distinguished is internal consistency from external separability. Consistency among many determinations establishes that no single instrument or pipeline is responsible for a result. It does not establish that the class of determinations, taken together, is free of a contribution common to all of them, because a common contribution is precisely what cross-validation among members of the class cannot detect.

Third, the question is not epistemically defeatist. It does not assert that cosmological knowledge is unreliable, that the measurements are wrong, or that the standard model of cosmology is mistaken. The claims of Chapters 4 and 5 are narrower and are stated so as not to imply any of these. Reliability and observer-independence are distinct properties, and a body of knowledge can possess the first in full measure while the second is not established.

The remainder of the paper answers the question in the negative and shows what follows.

3. Instruments and Foregrounds

Every measurement of the cosmic microwave background has been made from within the Solar System. COBE operated in low Earth orbit. WMAP and Planck operated at the second Sun-Earth Lagrange point, approximately 1.5 million kilometres from Earth. Ground-based and balloon-borne experiments operate at the Earth's surface or in its atmosphere. The most distant human-made object, Voyager 1, is at roughly 165 astronomical units, which is under one thousandth of the distance to the

nearest star; it carries no instrument capable of a microwave background measurement, and the observation has never been attempted from beyond the heliosphere.

The observations are therefore necessarily acquired from within the Solar System and are subject to line-of-sight foregrounds associated with the Solar System and with the Milky Way. These are of several distinct kinds and should not be conflated. Interplanetary dust produces thermal emission concentrated toward the ecliptic plane, which enters microwave measurements at a level that must be modelled. Galactic emission comprises synchrotron radiation, free-free emission, thermal dust and anomalous microwave emission, each with a characteristic frequency dependence and a distribution concentrated toward the galactic plane but present at all latitudes. Extragalactic point sources contribute at small angular scales. These components differ in physical origin, spectral behaviour and spatial distribution, and the degree of contamination varies strongly with direction and frequency.

Foreground separation is a mature discipline and its methods are effective. Multi-frequency observation exploits the different spectral signatures of the components; several independent algorithms are applied to the same data and their outputs compared; sky masks exclude the most contaminated regions and results are tested for stability as the mask is varied; different instruments with different systematics observe the same sky and their cleaned maps are compared. The consistency achieved by these procedures is real and is one of the reasons the cosmological parameters derived from the microwave background are as tightly constrained as they are.

The Planck analysis is the most developed instance and its structure is worth stating, because it shows how thorough internal validation can be. Four independent component-separation methods — Commander, NILC, SEVEM and SMICA — were applied to the same data, each resting on different assumptions about the foreground components and each producing its own cleaned map, and the results were compared against one another (Planck Collaboration 2020b). Residual foregrounds were characterised rather than assumed absent, and zodiacal emission from interplanetary dust was explicitly modelled and removed (Planck Collaboration 2014). The cosmological parameters derived from the resulting maps are consistent across temperature, polarization and lensing, and the base cosmological model fits them well (Planck Collaboration 2020c). None of this is in question here, and the argument does not require any of it to be revised.

What these procedures constitute is internal cross-validation. Their function is to establish that a result does not depend on any particular foreground model, algorithm, frequency channel, mask, or instrument. They perform that function well, and the claim that foreground removal cannot be validated would be false. What cannot be performed is an observationally external null test, which would require an observation of the same signal acquired under conditions in which the contaminating structures are differently oriented with respect to the observer. Every cleaned map is a map of the sky as seen from within the Solar System, at one epoch, by instruments constructed and calibrated within the domain whose data they produce. The distinction is between validating a model against other data sharing the observer's conditions and testing it against data that do not.

The distinction between these two things is not a criticism of the methods. It is a statement about what the methods establish. Constraining a foreground model within a set of observations sharing the same vantage is a different operation from confirming that the foreground model has removed everything associated with the vantage. The first is achievable and has been achieved. The second requires a measurement that has not been made.

The same distinction applies to the extension of instrumental range, and the point must be made carefully because the obvious objection arises here. Human perception responds to a narrow band of the electromagnetic spectrum. Instruments have removed this limitation comprehensively: radio, infrared, ultraviolet, X-ray and gamma-ray astronomy all register inputs to which no human sense responds, and the microwave background itself is invisible to unaided perception. An argument resting on the limits of human sensory bandwidth would therefore be refuted by any infrared telescope, and no such argument is made here.

What an instrument does is convert a physical input outside the range of direct perception into one inside it. The conversion is performed by a device built from materials of this domain, calibrated against standards of this domain, operating through the physical mechanism by which discrimination occurs in this domain, and read out by an observer of this domain. The instrument extends the range of inputs accessible within the domain. It multiplies channels. It does not manufacture an observer positioned outside the conditions under which observation occurs, because every component of the instrument, including the act of reading it, is subject to those conditions. Extending access within a domain and exiting a domain are operations of different kinds, and no accumulation of the first produces the second.

4. Large-Scale Alignment Anomalies

The lowest multipoles of the microwave background temperature field have been reported to exhibit features that are statistically unusual relative to statistical isotropy under the standard cosmological model. Statistical isotropy implies that the orientations of the multipole moments are random and mutually uncorrelated. Analyses of the first-year data from WMAP reported that the quadrupole and octopole are aligned with one another more closely than isotropy would suggest, and that their common axis lies near the ecliptic plane and near the direction of the dipole (de Oliveira-Costa et al. 2004); the configuration was subsequently given the informal name by which it is still known (Land & Magueijo 2005). A low quadrupole amplitude and a deficit of large-angle correlation had been reported earlier and were re-examined in the same period.

The subsequent history requires accuracy rather than summary, because the anomalies have not behaved uniformly. The alignment of the quadrupole and octopole was found in the WMAP nine-year data and again in Planck data, with orientations consistent between instruments that differ in frequency

coverage and systematics (Planck Collaboration 2020d). The assigned significance varies with the statistic chosen, with the treatment of the sky mask, and with whether the feature was specified before or after inspection of the data, and estimates in the literature therefore span a range rather than converging on a value (Schwarz et al. 2016). What has not occurred is the outcome that would have settled the matter: the features did not disappear as instruments improved, and they have not been attributed to a specific identified systematic. They remain listed among the unexplained large-angle results.

This chapter does not assert that the alignment is a property of the universe. It does not assert that the alignment is a residual of foreground emission. Both remain live, and the analysis here does not depend on either.

What is analysed is the structure of the indecision.

Three candidate explanations are in play. The first is that the alignment is a real large-angle feature of the cosmological signal, which would require a departure from statistical isotropy and would place a preferred direction in the largest-scale structure of the universe. The second is that the alignment is a residual of a foreground associated with the observer's location, and the association of the axis with the ecliptic is what makes this candidate natural, since the interplanetary dust distribution is organised by the same plane. The third is that the alignment is a statistical fluctuation. The quadrupole has five independent components and the octopole seven; with so few modes, unusual configurations occur with non-negligible probability, and the significance assigned to an alignment depends on which features are counted as anomalous and whether they were specified before or after inspection of the data.

The third candidate is limited by a structural constraint that has no remedy. There is one sky. The largest angular scales are sampled by a small number of modes, and no observation increases that number, because the number is set by the geometry of the observable region and not by the quality of the instrument. This is the cosmic variance limit, and at the lowest multipoles it was reached decades ago. Improving instruments cannot distinguish a real large-angle feature from a fluctuation at these scales, because the improvement adds precision to a measurement whose uncertainty is dominated by the finite number of available modes.

The first and second candidates are limited by a different constraint, and it is the one this paper is concerned with. They predict the same observation. A cosmological preferred direction that happens to lie near the ecliptic, and a foreground residual organised by the ecliptic, produce a signal with the same orientation. Domain-internal methods can constrain them. Frequency dependence, polarization data, cross-correlation with tracers of the foreground components, and the comparison of independently cleaned maps all bear on the question, and this paper does not claim that such methods are incapable of narrowing it; they have narrowed it. What they cannot supply is a test of a different kind. An observation of the same cosmological signal from a vantage where the contaminating structures are differently oriented with respect to the observer — a position sufficiently displaced from the Solar System that the interplanetary dust contribution is altered while the cosmological signal is not — would be qualitatively

different from any of them, because it would vary the observer's relation to the foreground rather than varying the model of it. No such observation exists. The relevant displacement is not the millions of kilometres separating existing spacecraft from Earth but a displacement comparable to the scale of the structure whose contribution is in question.

The indecision is therefore not a consequence of insufficient effort. Two decades of work by many groups, with successive instruments and independent pipelines, has clarified the anomalies without resolving them, and the reason is that the discriminating observation belongs to a class that domain-internal measurement does not contain. Internal cross-validation constrains foreground models; it cannot orient the observer differently with respect to the structures that generate the foreground.

The interest of the case for the present argument lies in one feature. The anomaly is an alignment between a cosmological measurement and the observer's own position. Under the standard reading, a coincidence between the largest-scale structure of the universe and the plane of a single planetary system is precisely the kind of thing that should not occur, and its occurrence is what makes the anomaly an anomaly. The reading given here does not require the coincidence to be real. It requires only the observation that the question of whether it is real is not decidable from within.

5. The Hubble Tension

The second case is of a different kind and the difference must be stated before the case is described, because conflating them would misrepresent both. The alignment anomalies concern a direction. The Hubble tension concerns a scalar. The expansion rate has no orientation, points nowhere on the sky, and bears no relation to the ecliptic or to any other plane. Any argument treating the two cases as instances of one phenomenon is mistaken.

The determinations divide into two classes by method rather than by instrument. The first infers the expansion rate from the microwave background: the angular scale of the acoustic features is measured with high precision, and converting that scale into an expansion rate requires the standard cosmological model to relate the sound horizon at recombination to the present epoch. The Planck analysis yields approximately 67.4 kilometres per second per megaparsec with an uncertainty near 0.5 (Planck Collaboration 2020c). The second measures the expansion rate directly through the distance ladder: geometric distances calibrate variable stars, variable stars calibrate supernovae, and supernovae reach the Hubble flow. The SH0ES analysis yields approximately 73.0 with an uncertainty near 1.0 (Riess et al. 2022). Neither determination is model-free: the first depends on the cosmological model relating the sound horizon to the present epoch, and the second depends on calibration and on astrophysical assumptions at each rung of the ladder. The distinction between them is not measurement against

inference but a model-dependent early-Universe inference against a late-Universe distance-ladder determination.

The discrepancy is around five standard deviations and has not diminished as measurements improved. Both sides have been examined intensively for systematic error over more than a decade without a resolution being found, and additional determinations have been brought to bear: time-delay strong lensing, megamaser distances to individual galaxies, the tip of the red giant branch as an alternative ladder calibration, and baryon acoustic oscillation measurements combined with independent constraints. These agree in some cases with one class and in some cases fall between, and their combination constitutes a network of cross-constraints of considerable complexity.

That network is real and effective, and the claim that domain-internal measurement cannot adjudicate the tension would be false. Internal adjudication is proceeding, it has ruled out many candidate explanations, and it may yet succeed in identifying a systematic error or in establishing that the discrepancy requires new physics. Nothing here asserts otherwise.

What holds is narrower and is not affected by whether the tension is resolved. Every determination in the network is obtained within one observing domain. The distance ladder is built from observations of luminous sources by instruments of this domain; the background inference uses measurements by instruments of this domain interpreted through a model constructed within it; the lensing, maser and oscillation determinations are the same in this respect. The network establishes that no single method carries the discrepancy. It cannot establish that the class of methods is free of a contribution common to all of them, because a common contribution does not appear as a disagreement among methods sharing it.

The structural interest of this case is different from the first and complements it. Here the discrepancy is between a quantity inferred through a model of the domain and a quantity determined by a ladder of local measurements within it. The model is the domain's account of itself: it describes the history, contents and geometry of the observable region, and it converts one measured angle into an expansion rate by that account. When the account and the local determination disagree, and the disagreement survives a decade of scrutiny, what is under strain is the relation between a domain's internal description of its own conditions and the readings taken under those conditions. The two cases therefore probe the same structure from different sides: one asks whether the observer's position can be separated from a directional measurement, the other whether the domain's self-description can be reconciled with a scalar reading taken inside it.

6. Spatial Displacement Is Not Ontological Displacement

The argument of this chapter rests on a result established in the third companion paper and inherited here rather than introduced as a definition. That result is the criterion of domain membership: an observing system belongs to a rendering domain in virtue of the rendering mechanism through which its discrimination is realised, and not in virtue of its spatial location or of the range of physical inputs its instruments can register. On that criterion, domains are individuated by mechanism rather than by position; they are therefore not spatial regions.

This matters for the form of what follows. If G were defined as the domain in which the observer happens to be, then the impossibility of leaving G would follow from the definition and would establish nothing. It is not so defined. G is determined by a criterion given independently in the companion paper, and the question of whether an observer can change G by changing position is therefore a substantive question with a determinate answer. The sequence is: the companion paper establishes the ontological criterion of domain membership; displacement alters the observer's state O ; whether it thereby alters G is what this chapter determines.

The general result is this. Given the inherited criterion of domain membership, an observer generated within a rendering domain cannot obtain an observation from outside that domain by changing position within it. Displacement changes O . It does not, by itself, alter G . The result is conditional on the criterion, and what it establishes is that spatial displacement is not a sufficient condition for a change of domain membership.

The argument has three parts, corresponding to the three ways in which an exit might be attempted: by moving, by building, or by inferring. Each is assessed against the inherited criterion, which is to say by asking whether the operation alters the mechanism through which discrimination is realised.

Consider movement first. A spacecraft carrying an instrument to another galaxy would occupy a position from which the Solar System's dust, the galactic disc, and the local structures now in the line of sight are differently arranged. Its measurements would differ from those made here, and the difference would be informative: it would constitute the external null test that Chapter 3 identified as absent, and it would settle Chapter 4's indecision. This is a genuine gain and the argument does not minimise it. But what has changed is the observer's position and the arrangement of contaminating structures with respect to it. The instrument aboard the spacecraft registers what instruments of this domain register, through the mechanism by which discrimination occurs in this domain, and its output is read by observers of this domain. The spacecraft carries the domain with it. It has moved within G ; it has not moved out of G . The distinction is not a matter of distance, and no distance affects it, because G is not a region.

Consider construction. An instrument responsive to physical inputs outside the range of any existing device would extend access further. The history of astronomy is a history of such extensions, and each has revealed structures that were inaccessible before. But an instrument is a physical object built from

materials of the domain, calibrated against standards of the domain, and read by a procedure belonging to the domain, and its response function is a physical property realised through the domain's mechanism of interaction. Its construction is an operation within the domain, and its output enters the domain's data. The number of channels increases; the conditions under which any channel is read do not change. As with movement, what accumulates is access, and access does not accumulate into exit.

Consider inference. It might be held that although no observation from outside is available, the conditions of the domain can be inferred from within by modelling them and subtracting their contribution. This is what foreground separation does, and what the dipole correction does, and the procedures work. But an inference of this kind establishes a decomposition only relative to the model performing it, and the model is constructed and validated within the domain against data that share the domain's conditions. To establish that the decomposition is unique, an external standard is required against which the residual could be checked, and Chapter 3 established what such a standard would have to be and that it does not exist. The absence is not of an unbuilt instrument but of a logically independent basis: there is no benchmark independent of the coupling between observer and domain against which a domain-conditioned datum could be uniquely resolved into an observer-independent remainder.

The three parts exhaust the ways in which the operation F of Chapter 2 might be realised. F does not exist within the domain.

Two consequences follow, and one misreading must be closed.

The first consequence is that observational reliability and observer-independence are distinct, and cosmology possesses the first without the second. This is not a deficiency in the practice. A body of measurements can be precise, mutually consistent, robust against systematic variation, and predictively successful, while remaining conditioned on the domain in which every one of them was obtained. The two properties are independent, and the second has been treated as though it followed from the first.

The second consequence concerns the Copernican principle. That principle states that observers occupy no special position, and it functions in cosmology as an assumption enabling the inference from what is observed here to what holds everywhere. Its status under the present result is not that it is false. Its status is that it cannot be tested by observers who cannot leave the domain, since any test would consist of comparing observations from different positions and every available position is inside. The principle is a condition of the inference, not a result of it.

The misreading to be closed is the one that has attended every position of this shape. The observer is the centre of the observer's own field of view. This is a geometric consequence of the finite speed of light and holds for any observer at any position: what can be seen is bounded by a surface centred on the observer, and that surface moves when the observer moves. It carries no implication whatever about the structure of what is observed. Rendering is not centred on the observer. The observer is itself rendered, along with the instruments and the structures observed, and the products of a rendering do not occupy privileged positions within it. The Earth is not the centre of the universe, the Solar System is

not a microcosm of the universe, and nothing in this paper asserts or requires either. What is asserted is that the field of view and the domain are two different things, that observers can move within both the first and the second, and that moving changes the first while leaving the second exactly as it was.

7. Consequences and Falsification

Two consequences require statement.

The first concerns how anomalies of the kind examined in Chapters 4 and 5 should be classified. In current practice an unresolved anomaly is a problem awaiting a solution, and effort is directed at improving the measurement until the ambiguity is removed. For anomalies whose candidate explanations differ in whether the effect belongs to the observer's conditions or to the world, improvement of measurement does not remove the ambiguity, because the candidates predict the same observation from every available vantage. Such anomalies are of a distinct kind and are misclassified when treated as ordinary open problems. Recognising the distinction has practical value: it identifies which questions further precision will settle and which require a different observation altogether.

The second concerns what would count as progress. The discriminating observations identified in this paper are specific rather than vague. A measurement of the microwave background from a position sufficiently displaced from the ecliptic plane and from the Solar System's dust distribution would separate the candidates of Chapter 4. Such a measurement is beyond present capability by a wide margin, but it is a measurement and not an impossibility, and stating it precisely is more useful than stating that the anomaly is hard.

The falsification conditions follow. Each is stated so that its satisfaction defeats the result rather than requiring its adjustment.

First, a demonstration that independent measurements from sufficiently separated locations uniquely separate the observer-position contribution from the cosmological signal. The result asserts that no domain-internal operation performs this separation uniquely. An explicit separation, exhibited and validated against a benchmark not itself constructed within the domain, ends the result. Separations relying on models built and checked within the domain do not satisfy this condition, and the condition is stated so as to exclude them.

Second, the construction of an instrument whose response is realised through a mechanism other than the rendering mechanism by which discrimination occurs in this domain, as that mechanism is characterised in the companion papers, and whose output can nevertheless be read by observers of this domain. The condition is operational: it requires exhibiting a registration process whose physical realisation falls outside that characterisation, together with a demonstration that its output enters this domain's data. The argument of Chapter 6 holds that instrumental extension multiplies channels within

a domain because every instrument is built, calibrated and read through the domain's mechanism. An instrument violating this while remaining readable would show that construction achieves what movement cannot.

Third, an identification of the origin of the alignment anomalies that is established without a measurement from a differently oriented vantage and is not model-dependent in the sense excluded by the first condition. The analysis of Chapter 4 holds that the candidates are not separable by domain-internal measurement. A separation achieved by other means would show the analysis to be wrong.

Fourth, a resolution of the Hubble tension by an independently established calibration or physical effect that demonstrably removes a common-mode contribution shared by the determinations. The claim of Chapter 5 concerns the class of determinations rather than its members, and it is not the claim that no such contribution can ever be identified. Identifying one, and showing by a route not itself dependent on the class that its removal reconciles the determinations, would defeat the claim as stated.

Fifth, an operational test of the Copernican principle that does not presuppose it. The result holds that the principle is a condition of cosmological inference rather than one of its findings, on the ground that every available observing position lies within one domain. A test satisfying the requirement would show the ground to be mistaken.

What remains to be said is the position itself, without qualification. An observer generated within a rendering domain does not leave that domain by moving within it. Instruments extend access inside a domain and do not create an observer outside one. A cosmological datum is conditioned on observer state, instrument and domain, and no operation available within the domain resolves it uniquely into an observer-independent quantity. Cosmology's measurements are reliable, mutually constraining, and conditioned. The first two properties have been established with great care over many decades. The third has not been examined. Observer dependence has a long history in physics, in quantum measurement theory, in relational formulations, and in operationalist accounts of physical quantities. What has not been treated as an explicit object of analysis is the observer's domain membership, which standard cosmological practice carries as an implicit condition of every datum it records.

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